

Green Home: A Reflection on the Relationship between Human and Furniture

Tianhui Huang
University College London
London, UK
ucjuth2@ucl.ac.uk

Tianming Ke
University College London
London, UK
ucjuexx@ucl.ac.uk

Yuyang Lin
University College London
London, UK
ucjuy04@ucl.ac.uk

Fangya Shi
University College London
London, UK
ucjufs1@ucl.ac.uk

Xinyu Zhang
University College London
London, UK
ucjuxz5@ucl.ac.uk

ABSTRACT

People nowadays tend to discard furniture and replace it frequently, paying less attention to daily maintenance or keeping it in good condition for a second-hand transaction. This is inconsistent with the UN's goal of responsible consumption and puts entire ecosystems and economies at risk. To solve this, we proposed Green Home, which encourages people to rethink their relationships and build an emotional connection with furniture through the analogy of plants. Our products consist of a plant-shaped tag with embedded sensors that can directly indicate the status of the furniture by colours, a wristband that provides a variety of interaction feedback to users, and an app that provides social and second-hand transaction functions. The difference between Green Home and existing solutions is that we focus on the entire furniture life cycle rather than just one part, achieving the UN's goal of responsible consumption through long-term emotional interaction.

CCS CONCEPTS

• **Human-centered computing** → **User centered design; User interface design; Interface design prototyping; User studies;** Haptic devices.

KEYWORDS

emotional connection, wearable, social media, internet of things

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1 INTRODUCTION

Every year, a large amount of intact furniture is left on the roadside and then sent to landfill and incineration, the figure for which

is increasing rapidly [5]. At the same time, according to World Furniture Outlook 2010, the rate of purchase and production of furniture is increasing each year dramatically [12]. Also, we found that people do not take much care of their furniture. It often happens that a lamp has fallen for days, or the sofa spilled with juice is not cleaned after several weeks. People are now used to short-lived furniture and inexpensive replacements [8].

Organisation for Economic Cooperation and Development [2002] defined sustainable consumption as, “*the [household] consumption of goods and services that meet basic needs and quality of life without jeopardising the needs of future generations*” (p. 16). Campbell believes that furniture is an important part of sustainable consumption, from its initial selection and purchase, through the entire process of use and maintenance, to its final disposal when it is no longer needed [2], and therefore Life Cycle Assessment (LCA) is increasingly seen as an important assessment method for sustainable consumption [9]. However, current solutions for sustainable consumption in furniture focus on a very limited part of the furniture life cycle, such as extending the service life at the product design level [1] and promoting second-hand trade as an alternative to disposal [6].

We want to design a product that extends the life of furniture at all stages of its life cycle. Our target users are likely to be broad, including anyone who is using furniture. However, we focus more on those 25–45 years old, who have a better acceptance of social media and a basic understanding of online second-hand transactions, while did not pay much attention to their furniture. We encourage people to rethink the relationship between themselves and furniture, helping them to build an emotional connection with their furniture so that they would indeed care about it from the moment they buy it, through the process they use it, to the moment they share it with the next owner. Furthermore, we also want to build social communities and facilitate credible second-hand transactions.

2 RESEARCH AND ESTABLISHING REQUIREMENTS

We immediately delved into exploring the behaviour patterns of participants' daily maintenance of furniture and their participation in second-hand furniture transactions. Our participants were from China, the UK, and other countries which would make our sample as representative as possible.

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First, we conducted a questionnaire to gather people's opinions preliminarily on our topic and received 264 valid responses. After analysing the data gathered, we have two significant findings: 68.6% of people agreed if the furniture were "alive", they would be more likely to treat them carefully; at the same time, 63.6% of people agreed that they would be more likely to participate in second-hand furniture transactions if they could know how the furniture was treated before by the previous owners. Thus, we can conclude that giving the furniture a "voice" to interact with humans can promote people to cherish furniture and participate in the second-hand trade more, which as a result would contribute to less furniture being discarded. Then we conducted semi-structured interviews with five individuals with a range of residence conditions and professions to gain a more in-depth and nuanced understanding of the topic.

An affinity diagram (Figure 1) was used to organise our data. Through brainstorming, organising, and summarising, we noticed participants spoke about their interest in the interaction with furniture in three aspects: the emotional connection with furniture, the novel design of the interaction system, and the extended interaction with friends. So, we focused on designing our products from the above three aspects.

These findings reinforced our choice of 25-45 years old who are open to new technology and social media, as our target group. This age group represents a significant portion of our participants and is prone to be more receptive to our ideas. We developed personas for this age group, including:

- Sarah, a 30-year-old UX designer who loves to share content on social media. She always pays attention to sustainable lifestyles, and she used to be troubled by how to dispose of some furniture after moving house.
- John, a 34-year-old engineer who is interested in new technology but is not very active on social media. He is concerned with sustainability but is not aware of the relationship between furniture and sustainable consumption.
- Emily, a 25-year-old graduate student who is active on social media and interested in novel interior designs. She is on a tight budget and often buys second-hand furniture from her friends or second-hand transaction platforms. She sometimes participates in volunteer activities and wants to help protect the environment.

These personas represent the different attitudes and behaviours of our target users towards furniture and technology.

Next, we created a list of core requirements for our product design:

- (1) The product must seamlessly integrate into day-to-day life and not disturb the daily use of furniture.
- (2) The product must not take a lot of time to learn how to use or put extra burdens on users.
- (3) The product should provide a social incentive to become a long-term companion to users.
- (4) The product should address daily maintenance as well as casual disposal issues.
- (5) The product should focus on building people's empathy for furniture.
- (6) The product should guide users to rethink the relationship between themselves and the furniture.



Figure 1: Affinity diagram and main findings

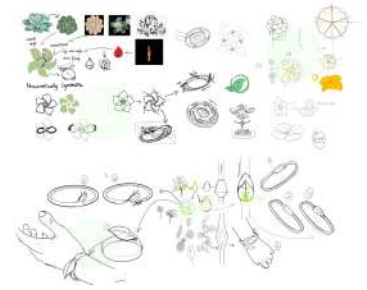


Figure 2: Initial sketches of the wristband and the tag

3 DESIGN PROCESS

After understanding the behaviour patterns of users, and being guided by the core requirements mentioned above, we had preliminary plans for the functionalities and forms of the product. We started by sketching out a variety of ideas and concepts. These sketches could depict different product forms and features, as well as possible interactions. After selecting from different sketches, we chose four that we thought exhibited the most promise. For each of these ideas, we next made storyboards that showed how the products would be used in various circumstances and how they would include the core requirements we had set. Here is a brief overview of these four plans.

- **VR Home** is built according to the real layout of the rooms, each furniture has been anthropomorphised, while the owner can check its status. The idea is inspired by the virtual tourism system by Wu et al. [2019], which used VR technology to enhance the emotional bond between people and environments [14].
- **IoT Community** aimed to integrate an IoT system in all pieces of furniture thus enabling people to see the status of furniture in real time. Besides, a digital community is set for them to join in and check their neighbour's furniture status. By comparing the condition of the furniture in everyone's home, people can realise the importance of protecting it.
- **Smart Glove** is integrated with multiple sensors, and it can be used to interact with furniture in the home. When the user touches the furniture with the glove, it can give instant feedback on the furniture status, such as scratches, to remind people of protecting it.
- **Band + Tag** connect users and furniture. Each piece of furniture is equipped with a smart tag, in the shape of a flower, to record the status changes during its service life. The user can use a wristband to make further connections with the furniture to obtain details of what it has experienced, voice, light, and vibration. The idea to use a plant metaphor is inspired by Hall [2022], as people can set up empathy through nurturing plants [7].

To reach the best solution, informal interviews were conducted. The interviews were conducted with the participants in the previous interviews. The participants were shown the preliminary storyboards during this session, and they were asked for feedback



Figure 3: Storyboard in family

Figure 4: Storyboard in community

Figure 5: Storyboard of feedback

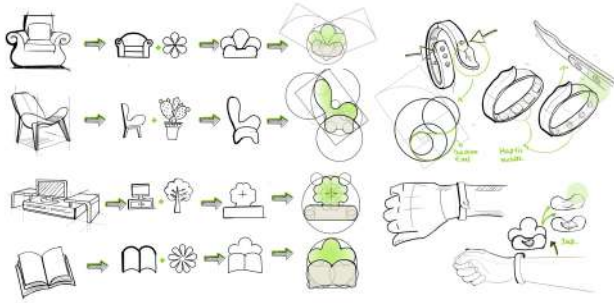


Figure 6: Final sketches

and opinions on the ideas. The goal of the interviews was to identify common attitudes among users to confirm the final plan. The interviews were informal in style and did not adhere to a predetermined script so that the participants could express their ideas freely. There were multiple pieces of feedback collected, but most were impressed by the band and IoT community. As many of them agreed that the VR room is attractive, but it is not possible for people to put on VR headsets to see the furniture all the time.

After the feedback was collected and analysed, we then decided to combine the band + tag with the IoT community. The functionality of community was adopted to build a long-term relationship between people and furniture, to inspire people's empathy, which can give people satisfaction when staying with the furniture [11]. This can build an emotional connection between people and furniture to the greatest extent possible. As the emotional connection is something that users cared about the most in the previous interviews, we paid more attention to this area in the final product, especially in the community interface more intimate and warmer.

At this stage, our members have confirmed the interface and focused functionalities of our product. We believed the final product interface will be simple and elegant in the shape, and not overwhelming in daily life. Besides, we explored many possible appearances of the tag and wristband, and tried to standardise their shapes with the golden ratio method (Figure 2). To finalise the functionalities, storyboards (Figure 3, 4, & 5) were also elaborated. Finally, we settled on 4 shapes of the tag, which were inspired by different types of furniture, as well as the wristband (Figure 6).

4 FINAL PRODUCT: GREEN HOME

Our product is inspired by people's attachment to plants. Using low-fidelity techniques, we came up with a set of devices and an

app to convey our concept of promoting responsible consumption throughout the life cycle of furniture use.

The devices (Figure 7) consist of two parts: a tag with an LED signal and a wearable wristband that has haptic simulators (Figure 8). The tag device which combined the shape of the furniture as well as a plant is made of liquid silica gel and cork which are both environmentally friendly materials. It should be first installed on the furniture as a symbol of the furniture with its LED signals indicating the health status of the furniture. Goldstein [1942] demonstrated that green and blue were considered to be relaxing, encourage an inward focus, and produce calm and stable action. So we chose green and blue for our light signal [4]. If the furniture was damaged by spilt liquid, serious hit, scrape, etc., the original green light which in total has five levels of brightness would dim (Figure 9). On the contrary, if the furniture was fixed timely after damage, the light would turn brighter. If the furniture was kept in a good state for a long time, the light would turn to a blue colour as a special incentive. Users then wear the wristband to connect to their furniture. The wristband in the shape of a leaf was taken from the intention of leaves coiled on the branches. When successfully connected, the wristband will provide vibration feedback, and the tag will show a breathing light effect. Human tactile perception is divided into three parts: the perception of vibration, the perception of pressure and the perception of temperature. Based on this, our haptic simulator consists of three main components, the vibration actuator, the pressure actuator and the thermoelectric actuator, the combination of which can simulate a wide range of complex tactile feedback [3]. If the furniture was split with liquid, users would feel cold water droplets. If the furniture was seriously hit, users would feel gathered pressure. If the furniture was scraped, users would feel pinpricks. In contrast, if the furniture was kept in good condition, users would feel like being softly tapped. At the same time, a voice broadcast would come out in the tone of furniture telling its own damage story. For example, a chair that was hit a few days ago might say: "Hey bro, you knocked me over when you threw your bag on me the other day, and I lay there all day, be careful next time!" With such feedback, people will empathise with the damage or care experienced by the furniture, leading to a closer emotional connection with them.



Figure 7: A full set of Green Home



Figure 8: Tactile simulators inside wristband

Our app is designed to give our users a whole control over the system. The app includes three main functions: checking the state of one's own furniture, visiting friends' furniture, and participating in second-hand transactions. When users log in, the first page would show the state of every piece of furniture connected to our "Green Home" system. People can also check the experience of

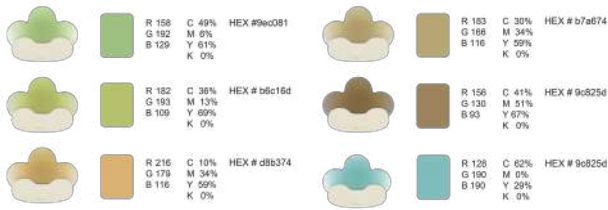


Figure 9: Different colours indicate different status



Figure 10: App pages for social and second-hand transaction functions

the furniture through our “review” function. Apart from taking care of their own furniture, users can also have a look at friends’ furniture. By visiting friends’ gardens, users can help to look after their friends’ furniture by sending messages like “You should have a look at your chair” to their friends. Or they could simply send likes or dislikes. Through this kind of social pressure, we hope to provide more motivation for users to cherish their furniture. In our second-hand market, each piece of furniture would carry a symbol showing the damage state of it for potential buyers to check. When users don’t need the furniture anymore, they can seek a new owner for their beloved furniture in our “Green Home” app (Figure 10).

5 USER TESTING

Using clay and paper with embedded sensors and LEDs, we created low-fidelity but functional physical prototypes, including both wristband and tag, and a high-fidelity, clickable app prototype with Figma (Figure 10). We invited 5 participants in London to experience all the functions of our product for 1.5 hours. We were interested in whether they could understand our entire ecosystem and whether they were able to build emotional connections during the whole process. This included understanding the plant-shaped tag on the furniture, using the wristband to touch the plant tag to get previous information and get multiple feedback, using our app and interacting with friends’ gardens, posting, and browsing second-hand transactions (Figure 11).

We conducted user tests based on the post-task walkthrough method, which means that the participants have to perform tasks in a recorded session and are then later asked by a moderator to reflect on their actions [13]. Each participant would go through the entire furniture life cycle from purchase and installation to daily use and transaction and would complete assigned tasks of specific scenarios, for example, a chair is accidentally fallen. After

completing the task, participants would watch the video recording and answer some simple questions. These questions were designed based on Jakob Nielsen’s 10 usability heuristic rules [10]. A sample of these questions is as follows:

- (1) Do you understand how this step was completed?
- (2) Do you understand why this feedback occurred?
- (3) Did this tactile feedback make you feel uncomfortable?
- (4) Do you have any further suggestions for the interaction to help build emotional connections?

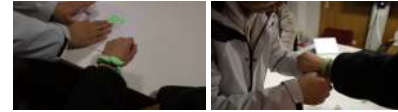


Figure 11: User testing of Green Home

After this one-and-a-half-hour experience, all five participants joined a 30-minute focus group discussion. One of the important findings is that users did not like the colour of the flower symbols since red seemed to release an alert signal. In addition, the leaf-shaped decoration on the band was suggested to fit more snugly on the wrist to avoid inconvenience, while it was noted that more positive feedback would make it easier to make an emotional connection.

Considering all the feedback, we refined our product and finally built a high-fidelity 3D model (Figure 7 & 9).

6 DISCUSSION

Green Home specifically highlights the emotional link between users and their furniture in a daily companionship pattern without pushing users to do any extra work. We consciously incorporated existing interaction patterns in both online and offline occasions and aimed to bring the users’ attention to the furniture’s feelings, which is beneficial to extend its “life” and achieve UN’s goal of responsible consumption. We delved into the entire life cycle of furniture use, integrating the concept of sustainable and responsible consumption into the whole ecosystem.

Our challenge during the user-centred design process was realising that not all types of human damage can be easily detected, and some people may never join our Green Home Community because of their solidified consumption views. From our first observation of the furniture discard problem to our literature review, then our surveys, affinity diagrams, and ideation, and finally the user-testing with our product, we adjusted our design solutions nearly in every step to make our final product more practical and user-friendly.

Our design proved to be useful during our usability testing with our target users, but we believe that it would be better to test the device in a real-life within-home environment, in order to observe the user’s natural behaviour. We could not completely replicate this environment during usability testing due to time and technique constraints.

One main function of the device that we want to validate further is the haptic feedback of making the users feel what the furniture has felt when damaged to determine whether it is too frustrating. We would also want to explore tracking users’ progress through

a website or mobile application. Ultimately, Green Home is a beneficial system for any user who wants to be a part of the new sustainable community.

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